Maths

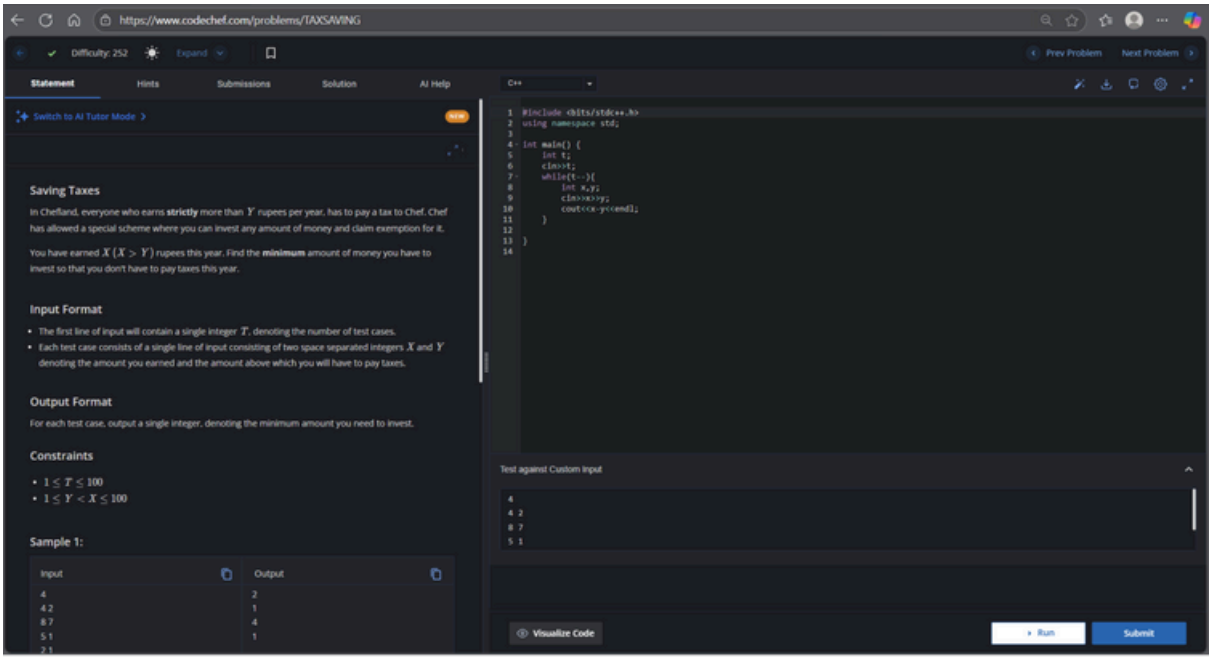
Name: Arjun Mankuskar

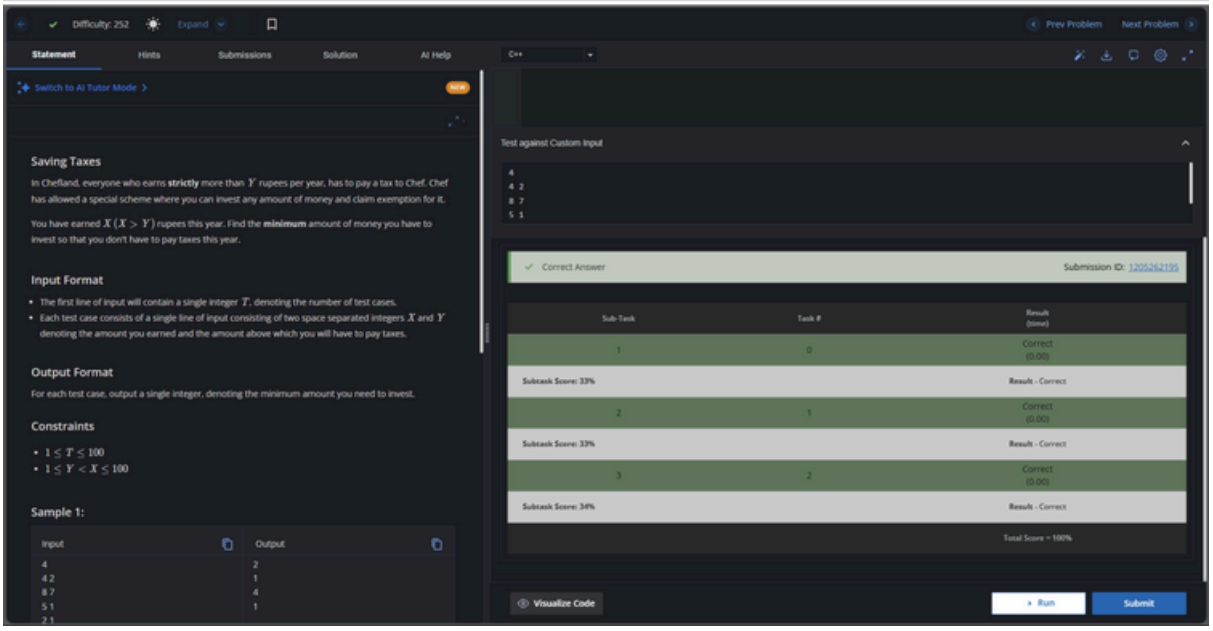
PRN:202401110038

Branch: CSE (AIML)

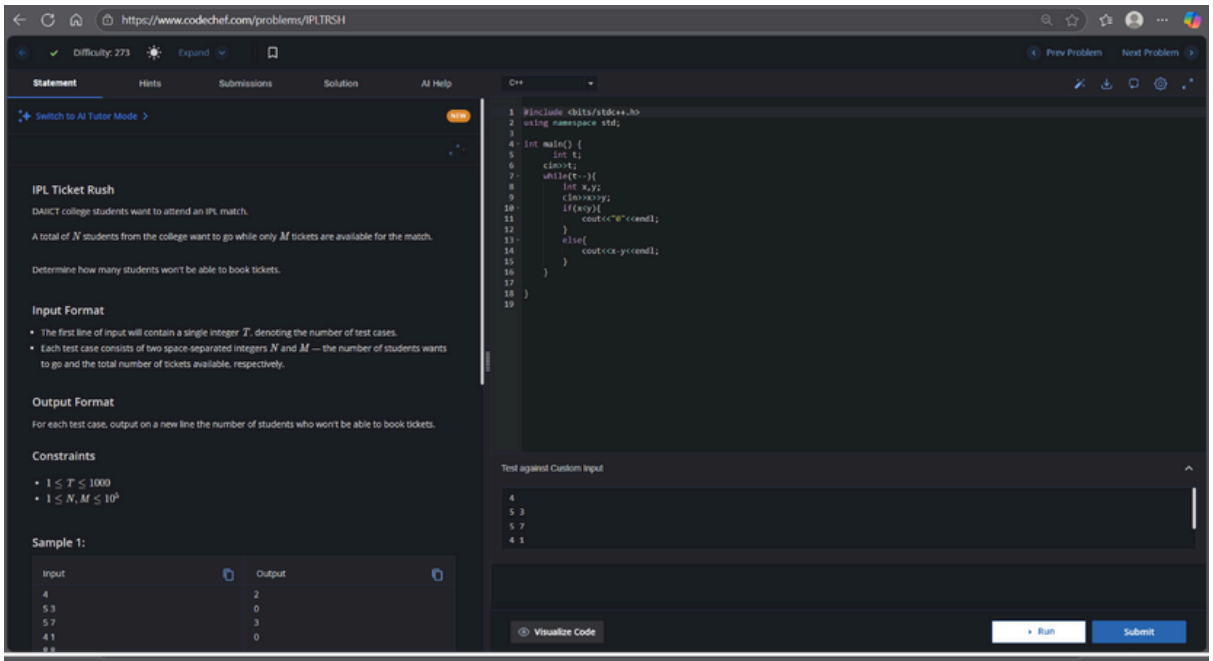
Batch: A2

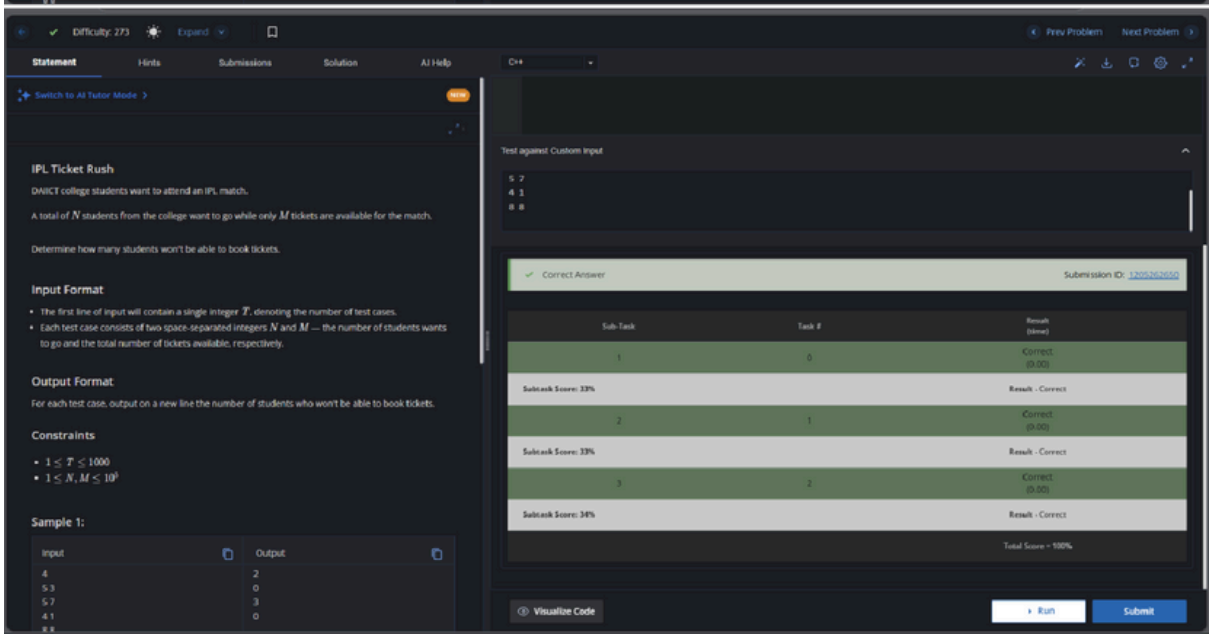
1.


https://www.codechef.com/problems/TAXSAVING
Difficulty: 252
Expand
Prev Problem
Next Problem
Statement
Hints
Submissions
Solution
AI Help
C++
Switch to AI Tutor Mode
Saving Taxes
In Chefland, everyone who earns strictly more than Y rupees per year, has to pay a tax to Chef. Chef has allowed a special scheme where you can invest any amount of money and claim exemption for it.
You have earned X ($X > Y$) rupees this year. Find the minimum amount of money you have to invest so that you don't have to pay taxes this year.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
Each test case consists of a single line of input consisting of two space separated integers X and Y denoting the amount you earned and the amount above which you will have to pay taxes.
Output Format
For each test case, output a single integer, denoting the minimum amount you need to invest.
Constraints
$1 \leq T \leq 100$
$1 \leq Y < X \leq 100$
Sample 1:
Input
Output
4
4 2
8 7
5 1
2 1
2
1
4
1
Test against Custom Input
Visualize Code
Run
Submit

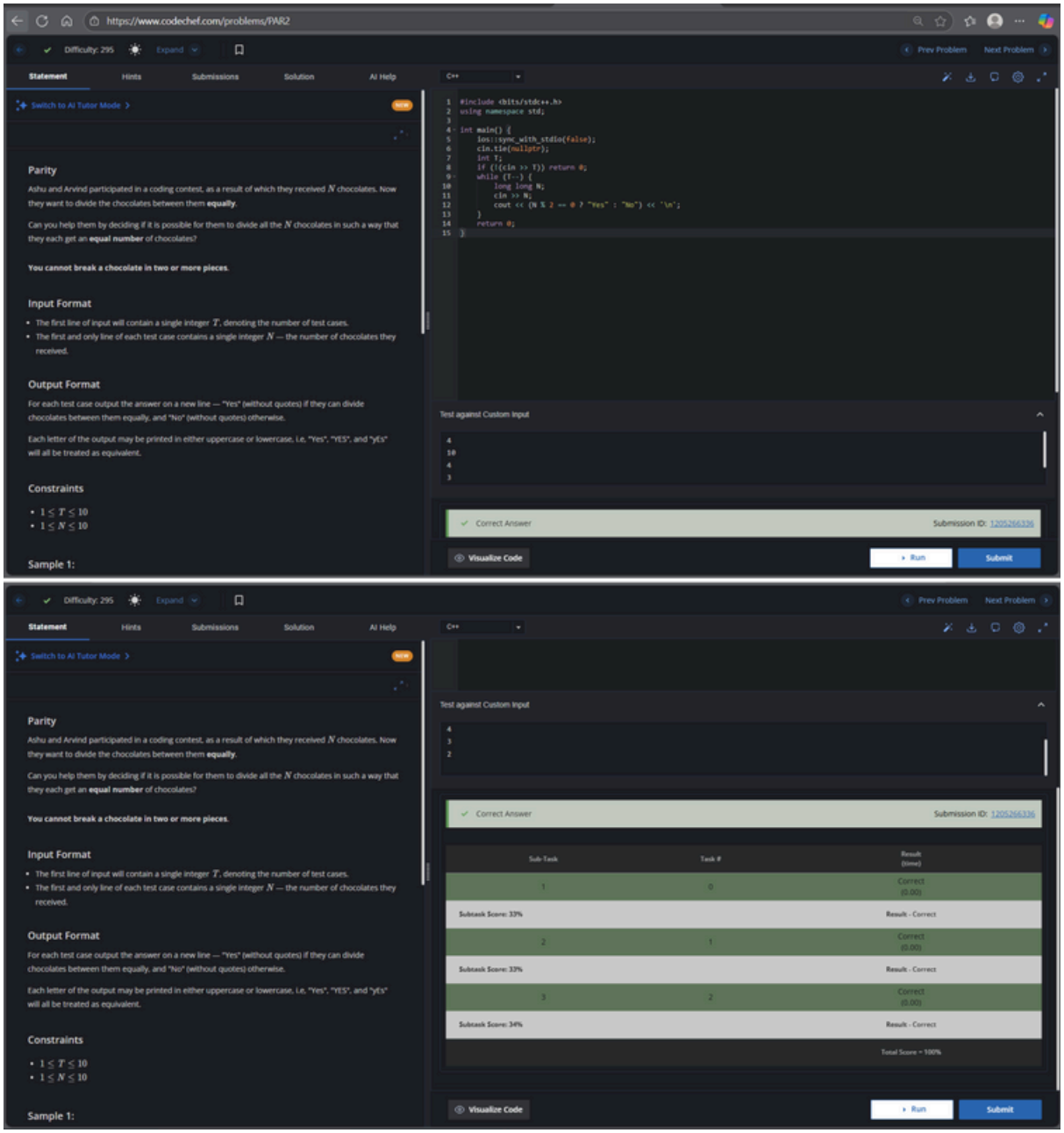

Difficulty: 252
Expand
Prev Problem
Next Problem
Statement
Hints
Submissions
Solution
AI Help
C++
Switch to AI Tutor Mode
Saving Taxes
In Chefland, everyone who earns strictly more than Y rupees per year, has to pay a tax to Chef. Chef has allowed a special scheme where you can invest any amount of money and claim exemption for it.
You have earned X ($X > Y$) rupees this year. Find the minimum amount of money you have to invest so that you don't have to pay taxes this year.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
Each test case consists of a single line of input consisting of two space separated integers X and Y denoting the amount you earned and the amount above which you will have to pay taxes.
Output Format
For each test case, output a single integer, denoting the minimum amount you need to invest.
Constraints
$1 \leq T \leq 100$
$1 \leq Y < X \leq 100$
Sample 1:
Input
Output
4
4 2
8 7
5 1
2 1
2
1
4
1
Test against Custom Input
Correct Answer
Submission ID: 1205262195
Sub-Task
Task #
Result (time)
1
0
Correct (0.00)
Subtask Score: 33%
Result - Correct
2
1
Correct (0.00)
Subtask Score: 33%
Result - Correct
3
2
Correct (0.00)
Subtask Score: 34%
Result - Correct
Total Score = 100%
Visualize Code
Run
Submit

2.


https://www.codechef.com/problems/IPLTRSH
Difficulty: 273
Expand
Prev Problem
Next Problem
Statement
Hints
Submissions
Solution
AI Help
Switch to AI Tutor Mode
C++
#include <bits/stdc++.h>
using namespace std;
int main() {
int t;
cin>>t;
while(t--){
int x,y;
cin>>x>>y;
if(x<y){
cout<<"0"<<endl;
}
else{
cout<<x-y<<endl;
}
}
}
IPL Ticket Rush
DAIICT college students want to attend an IPL match.
A total of N students from the college want to go while only M tickets are available for the match.
Determine how many students won't be able to book tickets.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
Each test case consists of two space-separated integers N and M — the number of students wants to go and the total number of tickets available, respectively.
Output Format
For each test case, output on a new line the number of students who won't be able to book tickets.
Constraints
1 ≤ T ≤ 1000
1 ≤ N, M ≤ 10^5
Sample 1:
Input
Output
4
5 3
5 7
4 1
2
0
3
0
Test against Custom Input
4
5 3
5 7
4 1
Visualize Code
Run
Submit


Difficulty: 273
Expand
Prev Problem
Next Problem
Statement
Hints
Submissions
Solution
AI Help
Switch to AI Tutor Mode
C++
IPL Ticket Rush
DAIICT college students want to attend an IPL match.
A total of N students from the college want to go while only M tickets are available for the match.
Determine how many students won't be able to book tickets.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
Each test case consists of two space-separated integers N and M — the number of students wants to go and the total number of tickets available, respectively.
Output Format
For each test case, output on a new line the number of students who won't be able to book tickets.
Constraints
1 ≤ T ≤ 1000
1 ≤ N, M ≤ 10^5
Sample 1:
Input
Output
4
5 3
5 7
4 1
2
0
3
0
Test against Custom Input
5 7
4 1
8 8
Correct Answer
Submission ID: 12052626
Sub-Task
Task #
Result (time)
1
0
Correct (0.00)
Subtask Score: 33%
Result - Correct
2
1
Correct (0.00)
Subtask Score: 33%
Result - Correct
3
2
Correct (0.00)
Subtask Score: 34%
Result - Correct
Total Score = 100%
Visualize Code
Run
Submit

3.


https://www.codechef.com/problems/PAR2
Difficulty: 295
Expand
Prev Problem
Next Problem
Statement
Hints
Submissions
Solution
AI Help
Switch to AI Tutor Mode
C++
#include <bits/stdc++.h>
using namespace std;
int main() {
ios::sync_with_stdio(false);
cin.tie(nullptr);
int T;
if (!(cin >> T)) return 0;
while (T--) {
long long N;
cin >> N;
cout << (N % 2 == 0 ? "Yes" : "No") << '\n';
}
return 0;
}
Parity
Ashu and Arvind participated in a coding contest, as a result of which they received N chocolates. Now they want to divide the chocolates between them equally.
Can you help them by deciding if it is possible for them to divide all the N chocolates in such a way that they each get an equal number of chocolates?
You cannot break a chocolate in two or more pieces.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
The first and only line of each test case contains a single integer N — the number of chocolates they received.
Output Format
For each test case output the answer on a new line — "Yes" (without quotes) if they can divide chocolates between them equally, and "No" (without quotes) otherwise.
Each letter of the output may be printed in either uppercase or lowercase, i.e. "Yes", "YES", and "yEs" will all be treated as equivalent.
Constraints
1 ≤ T ≤ 10
1 ≤ N ≤ 10
Sample 1:
Test against Custom Input
4
10
4
3
Correct Answer
Submission ID: 1205266336
Visualize Code
Run
Submit

Difficulty: 295
Expand
Prev Problem
Next Problem
Statement
Hints
Submissions
Solution
AI Help
Switch to AI Tutor Mode
C++
Parity
Ashu and Arvind participated in a coding contest, as a result of which they received N chocolates. Now they want to divide the chocolates between them equally.
Can you help them by deciding if it is possible for them to divide all the N chocolates in such a way that they each get an equal number of chocolates?
You cannot break a chocolate in two or more pieces.
Input Format
The first line of input will contain a single integer T, denoting the number of test cases.
The first and only line of each test case contains a single integer N — the number of chocolates they received.
Output Format
For each test case output the answer on a new line — "Yes" (without quotes) if they can divide chocolates between them equally, and "No" (without quotes) otherwise.
Each letter of the output may be printed in either uppercase or lowercase, i.e. "Yes", "YES", and "yEs" will all be treated as equivalent.
Constraints
1 ≤ T ≤ 10
1 ≤ N ≤ 10
Sample 1:
Test against Custom Input
4
3
2
Correct Answer
Submission ID: 1205266336
Sub-Task
Test #
Result (time)
1
0
Correct (0.00)
Subtask Score: 33%
Result - Correct
2
1
Correct (0.00)
Subtask Score: 33%
Result - Correct
3
2
Correct (0.00)
Subtask Score: 34%
Result - Correct
Total Score = 100%
Visualize Code
Run
Submit